

Birzeit University
Computer Science Department
Discrete Math (COMP 233)
Second Semester 2025/2026
Quiz #1: Chapters 1 + 2 + 3

Student Name: _____ Student ID: _____

Question #1: [Marks 45]

- a) Simplify the following logical expression to an equivalent expression: [15 pts]

$$\neg(p \wedge \neg q) \wedge (p \vee q)$$

- b) Show that the following expression is a Tautology by proving the logical equivalence using a truth table: [25 pts]

$$(p \wedge (p \rightarrow q)) \rightarrow q \equiv T$$

- c) Which Rule of Inference is represented by the tautology in **part (b)**? Write its formal argument form. [5 pts]

Question #2: [Marks 55]

- a) Write the **contrapositive**, **converse**, **inverse**, and **negation** of the following statement: [20 pts]

If Sara passes Discrete Math, then she will register for Data Structures and she will take Algorithms.

- b) Consider the following predicates defined on the domain $D = \{2, 3, 5, 6, 8\}$: [20 pts]

- $E(x)$: "x is even"
- $P(x)$: "x is prime"

For each of the following, write the statement **formally** using quantifiers and predicates, then determine its **truth value** (True or False). Justify your answer.

- i) *Every element in D is even.* [5 pts]
ii) *There exists an element in D that is both even and prime.* [5 pts]

Rewrite the following formal statement as a complete **English sentence**, then determine its **truth value** with justification: [10 pts]

$$\forall x \in D, (E(x) \rightarrow P(x))$$

- c) Provide the direct logical equivalent or operation for each of the following [15 pts]
(Write the formal expression):
- 1) Write the negation of the conditional statement: $p \rightarrow q$ [3 pts]
 - 2) Which variation of the conditional $p \rightarrow q$ is logically equivalent to its Converse? [3 pts]
 - 3) Given $p \leftrightarrow q$, write it as a conjunction of two conditional statements. [3 pts]
 - 4) Write the formal argument form of Modus Tollens (using p and q). [3 pts]
 - 5) Let $D = \{1, 2, 3, 4\}$ and let $P(x)$ be "x is a factor of 8." Write the truth set of $P(x)$. [3 pts]

Please, use the answer sheet ONLY!

Good Luck!